UNIVERSITY OF CAPE TOWN
IYUNIVESITHI YASEKAPA • UNIVERSITEIT VAN KAAPSTAD

EEE4022S

Final year project topics

2026

RA Verrinder

Topic list

- **RAV2026-01** BlueROV project: Stereo reconstruction of the sea ice underside
- **RAV2026-02** BlueROV project: Stereo visual-inertial odometry for under-ice navigation
- **RAV2026-03** SHARC project: Directional wave measurement across the ice-ocean transition
- **RAV2026-04** SHARC project: Inferring sea ice state and thickness from buoy inertial response
- **RAV2026-05** SnoCam+ project: Image-processing pipeline for sinking marine-snow particles
- **RAV2026-06** RGB-D microtopography of sandprawn bioturbation
- **RAV2026-07** Compact sealed stage-top incubator for long-duration live bacterial imaging

Note
If you are interested in selecting one of these topics, please **message me on MS Teams** if you need more information and use my **booking page** to book a meeting to discuss the project details.

MARiS Marine and Antarctic Research for Innovation and Sustainability
ARU AFRICAN ROBOTICS UNIT
ELECTRICAL ENGINEERING
SOCCO Southern Ocean Carbon-Climate Observatory
MMRU Molecular Mycobacteriology Research Unit

EEE4022S RAV2026-01

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-01**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	-	
Title	**BlueROV project: Stereo reconstruction of the sea ice underside**	

Description

Background

The distribution and behaviour of sea ice within the Marginal Ice Zone (MIZ) of the Southern Ocean strongly influence global climate processes, yet our understanding of the sea ice life cycle remains constrained by the limited availability of in situ observational data, particularly during the harsh Antarctic winter (**Vichi**, 2022). One of the least accessible surfaces in this system is the underside of the sea ice. Optical imaging beneath sea ice can reveal basal roughness, ice morphology, algae distribution, and the structure of the ice-ocean interface, all of which govern momentum and heat exchange and modulate the back-scatter signals used in satellite remote sensing (**Landy** et al., 2015, 2023; **Zhao** et al., 2022). Reconstructing the three-dimensional geometry of the ice underside is therefore a valuable but under-instrumented measurement, and under-ice operations are further constrained by limited GPS access, complex light conditions, and unstable navigation environments (**Zhao** et al., 2022).

The BlueROV2 (**Blue Robotics**, 2025) is a cost-effective research-class underwater Remotely Operated Vehicle (ROV) that has gained popularity in research communities due to its customisability, affordability, and ease of integration with external sensors. The UCT BlueROV2 includes a high quality stereo camera pair, a doppler velocity log (DVL), underwater illumination, an inertial measurement unit (IMU), a depth sensor, and side scan sonar, and provides feedback-stabilised six-degree-of-freedom motion from six T200 thrusters. A calibrated stereo camera pair is particularly well suited to surface reconstruction as it recovers metric scale directly from disparity, avoiding the scale ambiguity inherent in monocular reconstruction. By imaging an overhead surface and fusing multiple stereo views referenced to fixed external markers, a three-dimensional model of the surface can be recovered and its accuracy quantified against a target of known geometry.

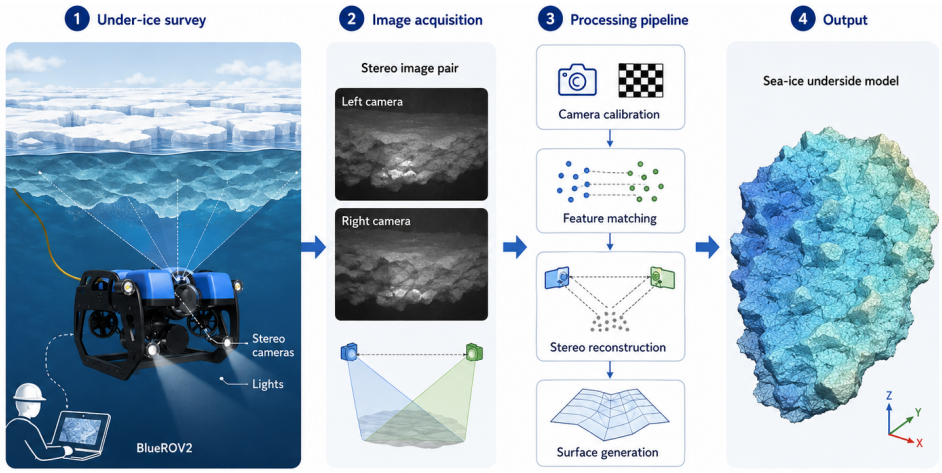


Fig. 1. Proposed stereo-reconstruction workflow for mapping the underside of sea ice. The BlueROV2 collects overlapping stereo images from below, which are calibrated, matched and fused to produce a metric 3D model of the ice underside. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation

Building on **Zhao** et al. (2022) and the surface characterisation work of **Spirakis** (2021); **Spirakis** et al. (2023); **Jones** (2024), this project develops and validates a stereo-camera three-dimensional reconstruction pipeline on the UCT BlueROV2 platform, as a first step towards reconstructing the underside of Antarctic sea ice. The work is conducted in a controlled pool environment, where a submerged proxy target of known geometry stands in for the ice underside. External fiducial markers provide the camera pose reference, so that the reconstruction is decoupled from onboard navigation, and the recovered surface model is evaluated against the ground-truth geometry of the proxy. The focus is on the geometric reconstruction itself: stereo calibration, handling of the underwater optical path, multi-view fusion, and quantified reconstruction accuracy.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current techniques and research that applies to this project. (c) To investigate underwater stereo reconstruction techniques, and select the most appropriate approach for the BlueROV2 platform. (d) To develop a test platform to collect stereo image data of a proxy surface in a controlled pool environment. (e) To develop and implement a stereo reconstruction pipeline that recovers a metric surface model referenced to external markers. (f) To test and evaluate the reconstruction accuracy against a target of known geometry and compare it to existing approaches. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements towards under-ice deployment. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A validated stereo reconstruction pipeline producing a metric surface model of a proxy target with characterised accuracy. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Image processing, 3D reconstruction, Interest in marine science
Extra Information	**References** **Blue Robotics**. BlueROV2. [ONLINE], 2025. Accessed: 2025-07-11. **url:** https://bluerobotics.com/store/rov/bluerov2/ D. **Jones**. SHARC project: Ice floe size and shape characterisation from LiDAR scans. BSc(Eng) final year project, Department of Electrical Engineering), University of Cape Town, 2024. **url:** https://tinyurl.com/djones2023 J. **Landy**, A. Komarov, and D. Barber. Numerical and experimental evaluation of terrestrial LiDAR for parameterizing centimeter-scale sea ice surface roughness. *IEEE Transactions on Geoscience and Remote Sensing*, 53: 4887–4898, 2015. **doi:** 10.1109/TGRS.2015.2412034 J. **Landy**, A. Komarov, and D. Barber. Secondary-scale surface roughness parameterization using terrestrial LiDAR. In *IGARSS 2015-2015 IEEE International Geoscience and Remote Sensing Symposium*, pages 2068–2017. IEEE, 2023. **doi:** 10.1109/igarss.2015.7326208 A. **Spirakis**. Investigation into surface roughness characterisation using LiDAR and cameras. BSc(Eng) final year project, Department of Electrical Engineering), University of Cape Town, 2021. **url:** https://tinyurl.com/spirakis2021 A. **Spirakis**, J. Hepworth, and R. Verrinder. 3D reconstruction of pancake sea ice using LiDAR and cameras. In *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 60–63. IEEE, 2023. **doi:** 10.1109/igarss52108.2023.10282250 M. **Vichi**. An indicator of sea ice variability for the Antarctic marginal ice zone. *The Cryosphere*, 16:4087–4106, 2022. **doi:** 10.5194/tc-16-4087-2022 L. **Zhao**, M. Zhou, and B. Loose. Towards under-ice sensing using a portable ROV. In *OCEANS 2022-Hampton Roads*, pages 1–8. IEEE, 2022. **doi:** 10.1109/OCEANS47191.2022.9977140
Broad Research Area	Underwater robotics, 3D reconstruction, Sea ice observation
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-01 GAs

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-01**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**-**	
Title	**BlueROV project: Stereo reconstruction of the sea ice underside**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the three-dimensional reconstruction of an overhead surface using a stereo camera pair on the BlueROV2 platform. It involves ill-posed and under-specified problems, such as the: • recovery of accurate surface geometry through the underwater optical path, • fusion of multiple stereo views into a single consistent metric model. It also presents high-level problems, such as: • Characterising the underside of sea ice, which is difficult to observe in situ. • Calibrating the stereo system and correcting for refraction and illumination underwater. • Referencing the reconstruction to external markers to decouple it from onboard navigation. The project also necessitates handling complex and incomplete information by: • Synthesising information from underwater imaging, photogrammetry, and sea ice literature. • Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Design a stereo data collection setup for a proxy surface in a controlled pool. • Develop and refine a stereo reconstruction pipeline referenced to external markers. • Validate the recovered surface model against a target of known geometry. • Compare reconstruction performance with existing approaches. • Apply statistical analysis to interpret reconstruction accuracy across trials.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational and experimental in nature and will require competencies in stereo calibration, image processing, 3D reconstruction, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate techniques and tools to achieve project objectives. **Software required may include:** • Python/C++ with OpenCV for stereo calibration and image processing • Photogrammetry or point cloud tools (e.g. COLMAP, Open3D) for reconstruction • MATLAB/Python for data processing, analysis, and validation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-01

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-02

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-02**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	-	
Title	**BlueROV project: Stereo visual-inertial odometry for under-ice navigation**	

Description

Background
Autonomous and remotely operated platforms are increasingly employed to extend our observational capabilities in harsh and inaccessible environments. In polar regions, under-ice exploration remains one of the most challenging and under-instrumented areas in marine science. Optical imaging beneath sea ice offers valuable insight into ice morphology, basal roughness, algae distribution, and ice-ocean interface processes, but under-ice operations are constrained by limited GPS access, complex light conditions, and unstable navigation environments (**Zhao** et al., 2022). Because satellite positioning is unavailable beneath the ice, and external acoustic positioning systems such as long baseline (LBL) or ultra-short baseline (USBL) are not viable, an under-ice vehicle must estimate its own trajectory from onboard sensing.

The BlueROV2 (**Blue Robotics**, 2025) is a cost-effective research-class underwater Remotely Operated Vehicle (ROV) that has gained popularity in research communities due to its customisability, affordability, and ease of integration with external sensors. The UCT BlueROV2 includes a high quality stereo camera pair, a doppler velocity log (DVL), underwater illumination, an inertial measurement unit (IMU), a depth sensor, and side scan sonar, and provides feedback-stabilised six-degree-of-freedom motion from six T200 thrusters. Visual-inertial odometry (VIO) combines onboard camera and inertial measurements to estimate the vehicle's trajectory and orientation in GPS-denied conditions (**Mourikis** and Roumeliotis, 2007; **Xu** et al., 2025). A calibrated stereo camera pair makes VIO considerably more tractable than a monocular approach, because metric scale is observed directly rather than being weakly recovered from inertial data alone, which is particularly valuable in the feature-poor conditions expected beneath sea ice.

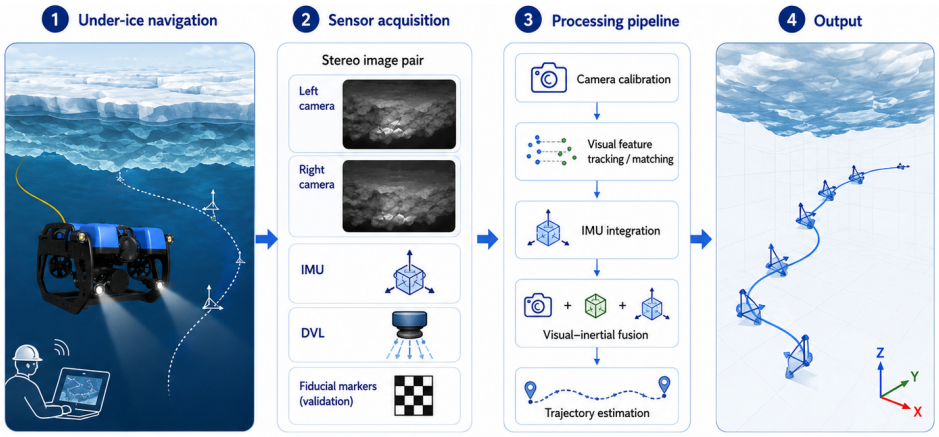


Fig. 2. Proposed stereo visual-inertial odometry workflow for under-ice navigation. Stereo camera, IMU and DVL measurements are combined to estimate the BlueROV2 trajectory, with fiducial markers used as an independent validation reference. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation
This project integrates and evaluates a stereo visual-inertial odometry pipeline on the UCT BlueROV2 platform in a controlled pool environment, as a first step towards robust navigation tools for under-ice applications (**Xu** et al., 2025; **Zhao** et al., 2022). The focus is on estimating the ROV's trajectory from the stereo camera pair and IMU, and on characterising localisation drift and accuracy across different motion profiles and visual conditions. The onboard DVL provides an independent dead-reckoned reference for validating the estimated trajectory, alongside external fiducial markers and a known survey path. As a further goal, the loose fusion of DVL velocity measurements into the estimator may be explored, as a route towards the fuller acoustic-visual-inertial systems required for reliable under-ice deployment (**Xu** et al., 2025).

Deliverables	**The key objectives of this project are:** (a) To understand the principles and challenges of visual-inertial odometry (VIO) in underwater GPS-denied environments. (b) To conduct a literature review of current VIO and SLAM techniques and evaluate their suitability for compact marine platforms such as the BlueROV2. (c) To integrate a stereo visual-inertial odometry pipeline onto the BlueROV2, including calibration and time synchronisation between the camera and inertial sensors. (d) To establish an independent ground-truth reference for validation using the DVL, external fiducial markers, and a known survey path. (e) To conduct controlled experiments in a pool environment to evaluate localisation drift, robustness, and accuracy under different motion profiles and visual conditions. (f) To critically analyse the system performance, draw conclusions about feasibility for under-ice deployment, and make recommendations. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A stereo visual-inertial odometry pipeline integrated on the BlueROV2 and tested in a controlled pool environment, with characterised drift and accuracy. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Image processing, Robotics, Interest in marine science
Extra Information	**References** **Blue Robotics**. BlueROV2. [ONLINE], 2025. Accessed: 2025-07-11. **url:** https://bluerobotics.com/store/rov/bluerov2/ A. I. **Mourikis** and S. I. Roumeliotis. A multi-state constraint Kalman filter for vision-aided inertial navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3565–3572. IEEE, 2007. **doi:** 10.1109/ROBOT.2007.364024 S. **Xu**, K. Zhang, and S. Wang. AQUA-SLAM: Tightly coupled underwater acoustic-visual-inertial SLAM with sensor calibration. *IEEE Transactions on Robotics*, 41:2785–2803, 2025. **doi:** 10.1109/tro.2025.3554396 L. **Zhao**, M. Zhou, and B. Loose. Towards under-ice sensing using a portable ROV. In *OCEANS 2022-Hampton Roads*, pages 1–8. IEEE, 2022. **doi:** 10.1109/OCEANS47191.2022.9977140
Broad Research Area	Underwater robotics, SLAM, Navigation
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-02 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-02**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**-**	
Title	**BlueROV project: Stereo visual-inertial odometry for under-ice navigation**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the integration of visual and inertial sensing for underwater odometry on the BlueROV2 platform. It involves ill-posed and under-specified problems, such as the: • development of sensor fusion software for underwater localisation. It also presents high-level problems, such as: • Maintaining localisation accuracy in underwater environments with limited visual features. • Calibrating the stereo camera and inertial sensors for time and spatial alignment. • Handling noise and drift in inertial sensors. • Deploying a testable system in a controlled pool environment as a proxy for under-ice conditions. The project also necessitates handling complex and incomplete information by: • Synthesising information from SLAM literature, underwater robotics research, and sensor specifications. • Balancing trade-offs between accuracy, computational load, and real-time feasibility.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Integrate a stereo visual-inertial odometry pipeline onto the BlueROV2 platform. • Develop and refine the calibration and time synchronisation between the camera and inertial sensors. • Establish an independent DVL and marker-based ground-truth reference. • Compare VIO performance across different motion profiles and visual conditions. • Apply statistical analysis to evaluate localisation accuracy and drift across trials.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational and experimental in nature and will require competencies in sensor data acquisition, state estimation, embedded programming, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate signal processing and data integration tools to achieve project objectives. **Software required may include:** • C++ and ROS for interfacing with the BlueROV2 and onboard sensors. • Python/MATLAB for data processing, calibration, and analysis. • OpenCV or similar libraries for visual feature tracking. • Stereo-inertial SLAM or state estimation frameworks (e.g. ORB-SLAM3, VINS-Fusion) for odometry.	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-02

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-03

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-03**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	-	
Title	**SHARC project: Directional wave measurement across the ice-ocean transition**	

Description

Background
Sea ice observation data are crucial to understanding climate perturbations, and the Marginal Ice Zone (MIZ) of the Southern Ocean, the transitional band between the open ocean and consolidated pack ice, is where much of this variability is set (**Vichi**, 2022). Waves are a primary driver: they flex, fracture, and melt the ice, and the ice in turn attenuates and reshapes the wave field (**Thomson** et al., 2021; **Hošeková** et al., 2020; **Ardhuin** et al., 2020). In situ wave measurement across the MIZ is therefore central to understanding wave-ice interaction, and the direction of the wave field, not only its height and period, governs how energy is delivered into the ice. The UCT SHARC buoy (**Noyce**, 2024; **Jacobson**, 2021) derives significant wave height and mean wave period from a vertical inertial time series, which captures how much the buoy heaves but neither which way the waves are travelling nor how faithfully the buoy is following them.

SHARC buoys are deployed onto the ice, and for much of a deployment they are ice-tethered. In this state the ice constrains the buoy, adding restoring stiffness and damping, so the buoy no longer follows the sea surface faithfully: its response to wave forcing is that of a low-pass filter, and higher-frequency wave motion is attenuated in the record. When the floe breaks up the buoy is released into open water and behaves closer to a free-floating wave follower, with a response amplitude operator (RAO) near unity across the wave band of interest. The floe may subsequently refreeze, returning the buoy to the ice-tethered state, so over a single deployment the buoy can transition repeatedly between the two regimes. Because the same measured motion maps to different true sea-surface motion in each regime, wave parameters derived under a single fixed assumption will be biased whenever the buoy is in the other state.

GPS-based measurement recovers the horizontal displacements of a buoy in addition to the vertical (**Herbers** et al., 2012), and the cross-spectra between the vertical and horizontal motions yield the directional Fourier coefficients of the wave field, from which a directional distribution can be reconstructed using estimators such as the truncated Fourier method (**Longuet-Higgins** et al., 1963) or the maximum entropy method (**Lygre** and Krogstad, 1986). Combining GPS horizontal displacements with the buoy's existing inertial heave therefore promises directional wave spectra on the SHARC platform, of the kind provided by directional drifting buoys in the open ocean (**Raghukumar** et al., 2019).

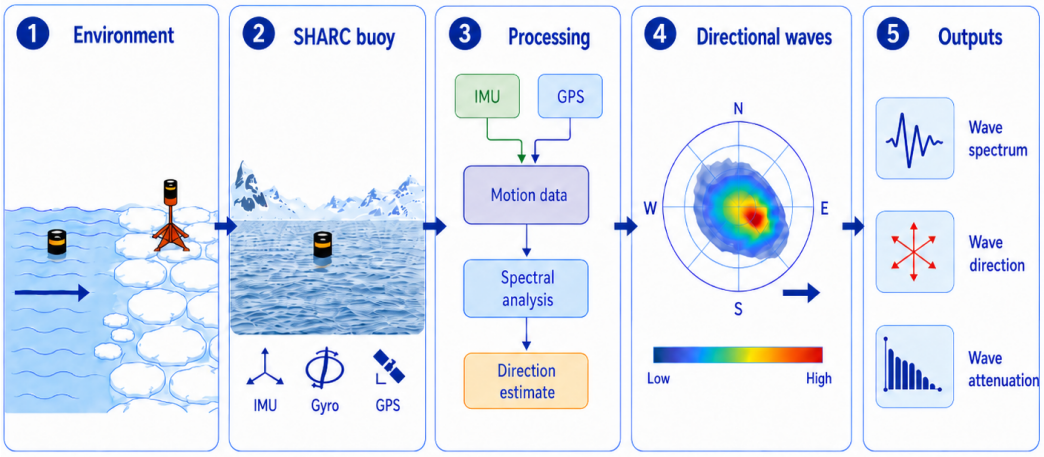


Fig. 3. Proposed SHARC processing workflow for directional wave measurement in the marginal ice zone. IMU, gyroscope and GPS motion data are analysed spectrally to estimate wave direction, wave energy and attenuation across the ice-ocean transition. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

The distinctive challenge is that a trustworthy measurement must hold across both regimes and detect which one it is in from the signal itself. The ice-tethered and open-water states leave distinct signatures in the inertial record: the spectral roll-off and its cut-off, the motion statistics, and the presence of transient floe impacts (**Noyce** et al., 2023). A pipeline that recognises the regime, applies the appropriate buoy response before estimating the wave field, and adapts as the buoy moves between states, can recover directional wave information consistently whether the buoy is filtering the field on the ice or following it in open water. The response of the buoy in each regime can be characterised directly (see also RAV2026-04, which identifies the buoy dynamics) or taken from prior work.

Subject of investigation
This project develops a regime-aware directional wave measurement pipeline for the SHARC buoy, approached as a signal-processing problem. The work: (i) estimates directional wave spectra from the combined GPS horizontal displacements and inertial heave; (ii) builds a detector that classifies the ice-tethered and open-water regimes, and the transitions between them including refreeze, from the spectral and statistical signatures of the inertial signal; and (iii) applies the appropriate buoy response in each regime, the low-pass response when ice-tethered and the near-unity RAO in open water, so that the wave field is recovered consistently in both states. The pipeline is validated in two complementary ways: against synthetic wave fields with imposed directional distributions and regime responses, where the true direction and regime are known and directional error and detection latency can be quantified, and against controlled wave-basin experiments, where physical waves of known direction are measured by the instrument.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of directional wave estimation and wave-ice interaction, and critically evaluate the techniques that apply to this project. (c) To investigate directional spectral estimation from combined GPS and inertial displacements, and select an approach suited to the SHARC buoy. (d) To develop a detector that identifies the ice-tethered and open-water regimes, and the transitions between them, from the inertial signal. (e) To develop and implement regime-aware processing that applies the appropriate buoy response in each state and recovers directional wave spectra across the transitions. (f) To validate the system against synthetic ground-truth fields and controlled wave-basin experiments, and characterise directional accuracy and detection performance. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements towards deployment. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A regime-aware directional wave measurement pipeline that detects the ice-tethered and open-water regimes and applies the appropriate buoy response in each, validated against synthetic and wave-basin data. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Signal processing, Spectral analysis, Interest in oceanography
Extra Information	**References** F. **Ardhuin**, M. Otero, S. Merrifield, A. Grouazel, and E. Terrill. Ice break-up controls dissipation of wind-waves across Southern Ocean sea ice. *Geophysical Research Letters*, 47:e2020GL087699, Apr 2020. **doi:** 10.1029/2020GL087699 T. H. C. **Herbers**, P. F. Jessen, T. T. Janssen, D. B. Colbert, and J. H. MacMahan. Observing ocean surface waves with GPS-tracked buoys. *Journal of Atmospheric and Oceanic Technology*, 29(7):944–959, Jul 2012. **doi:** 10.1175/JTECH-D-11-00128.1 L. **Hošeková**, M. P. Malila, W. E. Rogers, L. A. Roach, E. Eidam, and L. Rainville. Attenuation of ocean surface waves in pancake and frazil sea ice along the coast of the chukchi sea. *Journal of Geophysical Research: Oceans*, 125(12): e2020JC016746, Dec 2020. **doi:** 10.1029/2020JC016746 J. **Jacobson**. SHARC Buoy: Robust firmware design for a novel, low-cost autonomous platform for the Antarctic Marginal Ice Zone in the Southern Ocean. MSc thesis, Marine and Antarctic Research Cente for Innovation and Sustainability (MARiS), University of Cape Town, 2021. **url:** http://hdl.handle.net/11427/35788 M. S. **Longuet-Higgins**, D. E. Cartwright, and N. D. Smith. Observations of the directional spectrum of sea waves using the motions of a floating buoy. In *Ocean Wave Spectra*, pages 111–136. Prentice-Hall, 1963 A. **Lygre** and H. E. Krogstad. Maximum entropy estimation of the directional distribution in ocean wave spectra. *Journal of Physical Oceanography*, 16(12):2052–2060, Dec 1986. **doi:** 10.1175/1520-0485(1986)016<2052:MEEOTD>2.0.CO;2 M. **Noyce**, R. Verrinder, and M. Vichi. Identification of ice floe impacts in an inertial time series and implications for wave measurement in pancake ice. In *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 149–152. IEEE, 2023. **doi:** 10.1109/IGARSS52108.2023.10281906 M. **Noyce**. UCT SHARC Buoy V3.0: Waves-in-ice measurement in the Antarctic MIZ. MSc thesis, Marine and Antarctic Research Cente for Innovation and Sustainability (MARiS), University of Cape Town, 2024. **url:** http://hdl.handle.net/11427/40386 K. **Raghukumar**, G. Chang, F. Spada, C. Jones, T. Janssen, and A. Gans. Performance characteristics of "Spotter," a newly developed real-time wave measurement buoy. *Journal of Atmospheric and Oceanic Technology*, 36(6): 1127–1141, 2019. **doi:** 10.1175/JTECH-D-18-0151.1 J. **Thomson**, L. Hošeková, M. H. Meylan, A. L. Kohout, and N. Kumar. Spurious rollover of wave attenuation rates in sea ice caused by noise in field measurements. *Journal of Geophysical Research: Oceans*, 126:e2020JC016606, Mar 2021. **doi:** 10.1029/2020JC016606 M. **Vichi**. An indicator of sea ice variability for the Antarctic marginal ice zone. *The Cryosphere*, 16:4087–4106, 2022. **doi:** 10.5194/tc-16-4087-2022
Broad Research Area	Wave measurement, Directional spectra, Signal processing, Sea ice observation
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-03 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-03**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**-**	
Title	**SHARC project: Directional wave measurement across the ice-ocean transition**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the directional measurement of ocean waves from a buoy that operates in both ice-tethered and open-water regimes. It involves ill-posed and under-specified problems, such as the: • reconstruction of a directional distribution from a small number of noisy motion channels, which is inherently underdetermined, • detection of the ice-ocean regime and transition from a non-stationary signal. It also presents high-level problems, such as: • Estimating directional spectra from combined GPS and inertial displacements on the SHARC buoy. • Applying the appropriate buoy response in each regime, the low-pass response when ice-tethered and the near-unity RAO in open water, to recover the true wave field. • Adapting the estimation and quality control across the ice-tethered to open-water transitions, including refreeze, over which the signal statistics change. The project also necessitates handling complex and incomplete information by: • Synthesising information from directional wave estimation, wave-ice interaction, and sensor specifications. • Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Design a directional estimation pipeline from combined GPS and inertial displacements. • Develop and refine a detector for the ice-tethered and open-water regimes and the transitions between them. • Apply the appropriate buoy response in each regime and recover the wave field across the transitions. • Validate the directional estimates against synthetic ground-truth fields and controlled wave-basin experiments. • Compare the performance of the regime-aware system with a single-regime baseline. • Apply statistical analysis to interpret directional accuracy and detection performance across trials.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational and experimental in nature and will require competencies in spectral analysis, directional estimation, detection, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate signal processing tools to achieve project objectives. **Software required may include:** • MATLAB/Python for spectral analysis and directional estimation • MATLAB/Python for regime detection and change-point analysis • MATLAB/Python for synthetic wave field generation and validation • Data acquisition tools for the wave-basin experiments	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-03

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-04

Student Proposed?	Y/N	N/A
ID	**RAV2026-04**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr M Hillebrand (Department of Mathematics)**	
Title	**SHARC project: Inferring sea ice state and thickness from buoy inertial response**	

Description

The SHARC buoy (**Noyce**, 2024; **Jacobson**, 2021) measures its own motion using onboard inertial sensors, and this motion is used to infer the wave field. However, the buoy is not always a perfect wave follower, its response depends on what it is resting on. SHARC buoys are deployed onto sea ice, but the ice beneath the buoy is not fixed through time. At deployment the buoy may sit on a large consolidated plate. Later, after break-up, it may be supported by a much smaller floe, before entering open water or becoming trapped again during refreezing. Each support state gives the buoy a different dynamic response. This means that the same wave forcing can produce a different inertial signature depending on the state, and wave parameters derived using a single fixed response assumption will be biased whenever the buoy is in another regime (**Vichi**, 2022; **Thomson** et al., 2021).

On a large plate, where the ice extent is much greater than the flexural length, the buoy can be treated as a load on a floating elastic sheet (**Squire**, 2020). The plate flexes, rather than allowing the buoy to heave freely. As the flexural rigidity of the ice increases, and because this rigidity scales strongly with ice thickness, the plate introduces a low-pass response whose corner frequency is linked to the flexural length (**Meylan** et al., 2015). On a small floe, where the floe is much smaller than the flexural length, the ice does not flex in the same way. Instead, the floe behaves more like a rigid floating raft, with its own heave, pitch and roll resonances set by its waterplane area and inertia. In this case the buoy moves with the floe, and the resonant frequencies increase as the floe becomes smaller (**Meylan** and Squire, 1994). In open water, by contrast, the buoy behaves much more like a free wave follower, with a response amplitude operator (RAO) close to unity across the wave band of interest. These different states should leave recognisable signatures in the inertial record, including the ratio of pitch and roll energy to heave, the presence or absence of rigid-body resonance peaks, and the position of the high-frequency roll-off. In principle, the support state can therefore be inferred from the signal itself.

System identification provides a way to turn these signatures into a quantitative model of the buoy–ice response. The support state determines both the form of the model and the ice property that can be inferred from it. The inertial data collected during the SCALE Winter 2022 cruise, where SHARC buoys were deployed across the Antarctic MIZ in the manner of earlier in situ buoy campaigns (**Meylan** et al., 2014), include transient events such as floe impacts and energetic wave groups (**Noyce** et al., 2023). These events excite the system near its natural modes, while the continuous wave-driven motion provides a basis for estimating the frequency response. For a large plate, the fitted response can be used to estimate flexural rigidity and, with assumed ice properties, an approximate ice thickness (**Squire**, 1995). For a small floe, the response can be used to identify a rigid-body model and infer something about floe scale. For open water, the response provides the buoy-only hydrodynamic baseline. Since the buoy mass and geometry are known, these estimates are constrained by the physical system rather than being unconstrained curve fits.

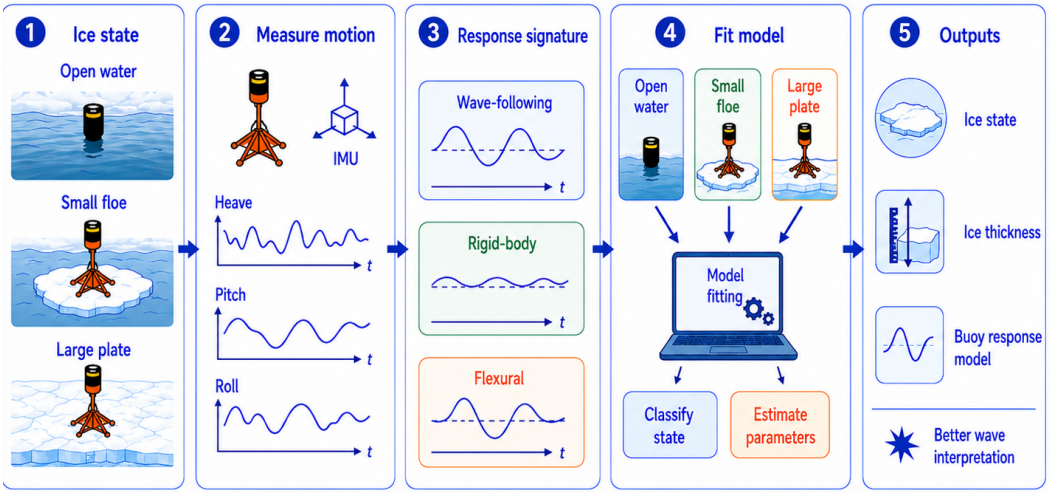


Fig. 4. Proposed method for inferring ice state from SHARC buoy motion. Heave, pitch and roll signals are used to distinguish open water, small-floe and large-plate conditions, and to fit the corresponding buoy-ice response model. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation

This project develops a method to infer, from the SHARC inertial signal alone, what the buoy is resting on and what dynamic response this introduces. The work uses the SCALE Winter 2022 data and focuses on the heave, pitch and roll response of the buoy. There are two main parts to the project. The first is a support-state detector that classifies the buoy as being in open water, on a small rigid floe, or on a large flexural plate, and tracks how this state changes through a deployment. This classification is based on inertial signatures such as the pitch/roll to heave energy ratio, the presence of rigid-body resonance peaks, and the form of the high-frequency roll-off.

The second part is the identification of the dynamic response within each state. For a large plate, the flexural response is fitted to estimate the effective flexural rigidity and, using assumed ice properties, an approximate ice thickness. For a small floe, a rigid-body model is identified and its resonances are related to floe scale. In open water, the buoy response is used to confirm the near-unity RAO and establish the buoy-only baseline. The response models are identified using free-decay realisations from transient events, cross-checked against frequency responses estimated from the continuous wave-driven motion.

The method is validated first against synthetic inertial signals, where the true support state and model parameters are known, allowing classification accuracy and parameter error to be quantified. Where suitable independent ice observations are available from the cruise, the inferred ice properties are also compared against measured conditions. The resulting state classification and response models provide the information needed by the wave-measurement pipeline to recover the wave field consistently across a deployment, rather than applying one fixed buoy response to all ice and open-water conditions (see also RAV2026-03).

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of system identification, floating-body and floating elastic-plate dynamics, and wave-ice interaction, and critically evaluate the identification techniques that apply to this project. (c) To extract, condition, and segment the SCALE Winter 2022 inertial data, and identify transient response events and continuous wave-driven segments suitable for analysis. (d) To develop a detector that classifies the support as a large flexural plate, a small rigid floe, or open water, and tracks the transitions between these states, from the inertial signatures. (e) To identify the dynamic response in each state by free-decay realisation cross-checked against a frequency response estimated from the wave-driven motion. (f) To estimate the ice parameters per state: the effective flexural rigidity, and hence an approximate thickness, on a large plate; the rigid-body model and floe scale on a small floe; with the open-water response as the buoy-only baseline. (g) To validate the classifier and parameter estimates against synthetic signals with known states and parameters, and, where available, against independent ice observations from the cruise. (h) To discuss performance, draw conclusions, and make recommendations for future improvements, such as response correction for improved wave measurement and ice characterisation from buoy motion. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A method that classifies the buoy's support as a large flexural plate, a small rigid floe, or open water from the inertial signal, together with the regime-specific dynamic models and ice parameter estimates, comprising the effective flexural rigidity and approximate thickness on a large plate, the rigid-body model and floe scale on a small floe, and the buoy-only baseline in open water, identified from the SCALE Winter 2022 data. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	System identification, Signal processing, Control, Interest in oceanography
Extra Information	**References** J. **Jacobson**. SHARC Buoy: Robust firmware design for a novel, low-cost autonomous platform for the Antarctic Marginal Ice Zone in the Southern Ocean. MSc thesis, Marine and Antarctic Research Cente for Innovation and Sustainability (MARiS), University of Cape Town, 2021. **url:** http://hdl.handle.net/11427/35788 M. H. **Meylan** and V. A. Squire. The response of ice floes to ocean waves. *Journal of Geophysical Research: Oceans*, 99(C1):891–900, 1994. **doi:** 10.1029/93JC02695 M. H. **Meylan**, L. G. Bennetts, and A. L. Kohout. In situ measurements and analysis of ocean waves in the Antarctic marginal ice zone. *Geophysical Research Letters*, 41(14):5046–5051, 2014. **doi:** 10.1002/2014GL060809 M. H. **Meylan**, L. G. Bennetts, C. Cavaliere, A. Alberello, and A. Toffoli. Experimental and theoretical models of wave-induced flexure of a sea ice floe. *Physics of Fluids*, 27(4):041704, Apr 2015. **doi:** 10.1063/1.4916573 M. **Noyce**, R. Verrinder, and M. Vichi. Identification of ice floe impacts in an inertial time series and implications for wave measurement in pancake ice. In *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 149–152. IEEE, 2023. **doi:** 10.1109/IGARSS52108.2023.10281906 M. **Noyce**. UCT SHARC Buoy V3.0: Waves-in-ice measurement in the Antarctic MIZ. MSc thesis, Marine and Antarctic Research Cente for Innovation and Sustainability (MARiS), University of Cape Town, 2024. **url:** http://hdl.handle.net/11427/40386 V. A. **Squire**. Geophysical and oceanographic information in the marginal ice zone from ocean wave measurements. *Journal of Geophysical Research: Oceans*, 100(C1):997–998, 1995. **doi:** 10.1029/94JC02813 V. A. **Squire**. Ocean wave interactions with sea ice: a reappraisal. *Annual Review of Fluid Mechanics*, 52:37–60, Jan 2020. **doi:** 10.1146/annurev-fluid-010719-060301 J. **Thomson**, L. Hošeková, M. H. Meylan, A. L. Kohout, and N. Kumar. Spurious rollover of wave attenuation rates in sea ice caused by noise in field measurements. *Journal of Geophysical Research: Oceans*, 126:e2020JC016606, Mar 2021. **doi:** 10.1029/2020JC016606 M. **Vichi**. An indicator of sea ice variability for the Antarctic marginal ice zone. *The Cryosphere*, 16:4087–4106, 2022. **doi:** 10.5194/tc-16-4087-2022
Broad Research Area	System identification, Buoy and ice dynamics, Sea ice thickness, Sea ice observation
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-04 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-04**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr M Hillebrand (Department of Mathematics)**	
Title	**SHARC project: Inferring sea ice state and thickness from buoy inertial response**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in determining, from field inertial data, what a SHARC buoy is resting on and identifying the corresponding ice model. It involves ill-posed and under-specified problems, such as the: • identification of a model from data in which the wave forcing is not directly measured, • selection of the model structure itself, since the support, a large flexural plate, a small rigid floe, or open water, is not known in advance and must be inferred from the signal, • estimation of ice parameters such as flexural rigidity and thickness from a response that only partially constrains them. It also presents high-level problems, such as: • Extracting and conditioning suitable transient and wave-driven segments from the SCALE Winter 2022 inertial series. • Discriminating the large-plate, small-floe, and open-water states from inertial signatures, and tracking the transitions between them. • Fitting the appropriate flexural or rigid-body model in each state, constrained by the known buoy properties. • Relating the identified response to ice properties, the flexural rigidity and thickness on a plate and the floe scale on a small floe. The project also necessitates handling complex and incomplete information by: • Synthesising information from system identification, floating-body and elastic-plate dynamics, and wave-ice literature. • Balancing technical challenges, engineering design, and stakeholder interests.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Extract, condition, and segment the SCALE Winter 2022 inertial data for analysis. • Develop and refine a detector for the large-plate, small-floe, and open-water states from inertial signatures. • Identify the dynamic response in each state by free-decay realisation and frequency-response estimation. • Estimate the ice parameters per state, the flexural rigidity and approximate thickness on a plate and the floe scale on a small floe. • Validate the classifier and parameter estimates against synthetic signals and, where available, independent ice observations. • Apply statistical analysis to interpret classification accuracy and parameter error across states and trials.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational in nature and will require competencies in data conditioning, system identification, spectral analysis, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate identification and signal processing tools to achieve project objectives. **Software required may include:** • MATLAB/Python for data conditioning and signal processing • MATLAB (System Identification Toolbox) or Python for system identification and frequency-response estimation • MATLAB/Python for support-state classification and synthetic signal generation • MATLAB/Python for model validation and analysis	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-04

Question	Question Text	Yes	No
Q1	Does this project involve data collection		X
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-05

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-05**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr S Thomalla (CSIR SOCCO)**	
Title	**SnoCam+ project: Image-processing pipeline for sinking marine-snow particles**	

Description

Background
Sinking particles, the marine snow of aggregates, faecal pellets, and detritus that settle out of the surface ocean, drive the biological carbon pump and transfer of the order of a gigatonne of carbon to the deep ocean each year (**Durkin** et al., 2021). Quantifying this flux, and the different particle types that carry it, is central to understanding ocean carbon sequestration, and the Southern Ocean is a globally important but under-observed region for it (**CSIR**, 2025). Traditional sediment-trap and Marine Snow Catcher methods capture particles for later analysis, but they integrate over time and demand labour-intensive manual microscopy. In situ and trap-deployable imaging systems, which image settling particles directly, offer a route to resolving particle flux and composition at far finer temporal resolution (**Song** et al., 2025; **Edirisinghe** et al., 2024).

Extracting quantitative flux from such imagery is fundamentally an image-processing problem. The particles are low in contrast and often transparent, the illumination varies, with a day-night cycle in surface deployments, and the imaging surface and its surroundings clutter the field of view. In the SnoCam+ configuration the particles settle onto a plate and accumulate over time, so a processing pipeline must remove the varying illumination and the growing background of already-settled material, segment individual particles from noise even as they begin to overlap, detect which particles are newly arrived between successive frames so that each is counted once, classify them into ecologically meaningful types, and convert particle size and shape into a particulate organic carbon estimate (**Song** et al., 2025; **Durkin** et al., 2021). Recent open-source systems such as GelCam (**Song** et al., 2025) and SnoCam+ (**Edirisinghe** et al., 2024) have demonstrated this workflow, but the classification stage in particular has largely relied on manual identification, and automating it remains an open challenge (**Amaral** and Durkin, 2025).

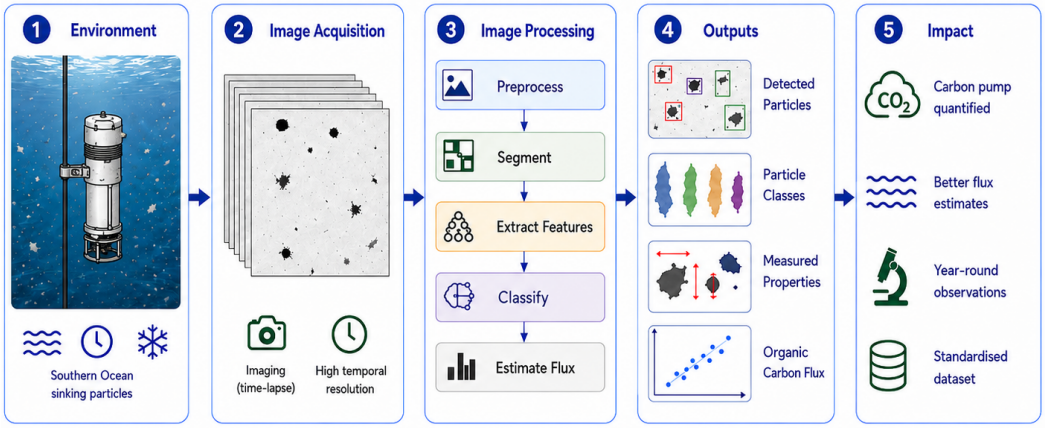


Fig. 5. Proposed image-processing workflow for sinking marine-snow particles. Time-lapse images are preprocessed, segmented and classified to measure particle properties and estimate particulate organic carbon flux. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation
This project develops an end-to-end image-processing pipeline for sinking marine-snow particles, in collaboration with the Southern Ocean Carbon-Climate Observatory (SOCCO). Building on the GelCam (**Song** et al., 2025) and SnoCam+ (**Edirisinghe** et al., 2024) workflows, the pipeline will preprocess the imagery to remove varying illumination and the accumulating background, segment individual particles, detect which particles are newly settled between successive frames so that each is counted once, classify them into particle types, and estimate the particulate organic carbon flux from the arrival rate and established size-to-carbon relationships (**Durkin** et al., 2021). The classification stage will use morphological features, such as size, shape, and intensity, with a classical classifier, following the manual scheme of **Durkin** et al. (2021); **Song** et al. (2025), with a machine-learning classifier identified as a route for future work once a sufficient labelled dataset is available (**Amaral** and Durkin, 2025). As reference imagery with known particle counts and composition may not be available, the pipeline is validated primarily against synthetic image sequences, generated with prescribed particle size, shape, and arrival-rate distributions on a modelled plate with a growing background, so that the segmentation, new-arrival detection, classification, and recovered carbon flux can be quantified against a ground truth that is known by construction. Where sinking-particle imagery is available through the SOCCO collaboration or published reference datasets, it is used to test the pipeline on real data.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of sinking-particle imaging and image-processing pipelines, and critically evaluate the techniques that apply to this project. (c) To develop preprocessing that removes the varying illumination and static background from the imagery. (d) To develop segmentation and new-arrival detection that identify individual particles and establish their arrival rate against the accumulating background. (e) To develop a morphological classification of particles into ecologically meaningful types. (f) To estimate the particulate organic carbon flux from the classified particles and their arrival rate. (g) To generate synthetic image sequences with known particle size, shape, and arrival-rate distributions as a ground-truth benchmark, and validate the pipeline against them, supplemented by reference imagery where available. (h) To discuss the performance of the pipeline, draw conclusions, and make recommendations for future improvements, such as machine-learning classification and onboard processing. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) An end-to-end image-processing pipeline that segments, detects newly settled particles, classifies, and estimates the carbon flux of sinking marine-snow particles from imagery. (d) A synthetic marine-snow image generator with a known ground truth, and a quantified validation of the pipeline against it and against reference imagery where available. (e) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (f) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (g) A poster summarising the research project report. (h) A seminar to the Department summarising the research project report. (i) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Image processing, Signal processing, Interest in marine science
Extra Information	**References** V. J. **Amaral** and C. A. Durkin. A computer vision-based approach for estimating carbon fluxes from sinking particles in the ocean. *Limnology and Oceanography: Methods*, 23(2):117–130, 2025. **doi:** 10.1002/lom3.10665 C. A. **Durkin**, K. O. Buesseler, I. Cetinić, M. L. Estapa, R. P. Kelly, and M. Omand. A visual tour of carbon export by sinking particles. *Global Biogeochemical Cycles*, 35(10):e2021GB006985, 2021. **doi:** 10.1029/2021GB006985 S. **Edirisinghe**, Y. Song, P. Roberts, et al. SnoCam+: An open-source, trap-deployable imaging system for sinking particles in the ocean. In *2024 Ocean Sciences Meeting*. American Geophysical Union, 2024 **CSIR**. Southern Ocean Carbon Climate Observatory (SOCCO). [ONLINE], 2025. Accessed: 2025-07-11. **url:** https://socco.org.za/ Y. **Song**, M. Omand, C. A. Durkin, M. L. Estapa, and K. O. Buesseler. GelCam: Visualizing sinking particle flux via a polyacrylamide gel-based sediment trap. *Limnology and Oceanography: Methods*, 23:715–728, 2025. **doi:** 10.1002/lom3.10724
Broad Research Area	Image processing, Particle imaging, Ocean carbon flux
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-05 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-05**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr S Thomalla (CSIR SOCCO)**	
Title	**SnoCam+ project: Image-processing pipeline for sinking marine-snow particles**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the automated extraction of sinking-particle flux from image sequences. It involves ill-posed and under-specified problems, such as the: • segmentation of low-contrast and transparent particles from a varying, cluttered, and accumulating background, • classification of particles into ecologically meaningful types from limited image information. It also presents high-level problems, such as: • Removing the varying illumination and static trap superstructure from the imagery. • Detecting newly settled particles between frames, against a background of already-settled material, so that each is counted once. • Converting two-dimensional particle images into a particulate organic carbon estimate. The project also necessitates handling complex and incomplete information by: • Synthesising information from image processing, sinking-particle imaging, and ocean carbon literature. • Validating the pipeline in the absence of guaranteed reference imagery, by constructing synthetic sequences with a known ground truth. • Balancing technical challenges, engineering design, and the interests of the SOCCO collaboration.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Develop and refine preprocessing to remove varying illumination and static background. • Develop segmentation and new-arrival detection of individual particles against the accumulating background. • Develop and refine a morphological classification of particle types. • Estimate the carbon flux from the classified particles and their arrival rate. • Generate synthetic image sequences with a known ground truth and validate the pipeline against them, supplemented by reference imagery where available. • Apply statistical analysis to evaluate the pipeline performance across the dataset.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational in nature and will require competencies in image processing, feature extraction, classification, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate image processing and data analysis tools to achieve project objectives. **Software required may include:** • Python/MATLAB with OpenCV or scikit-image for image processing and segmentation • Python/MATLAB for new-arrival detection and feature extraction • Python/MATLAB for synthetic particle-image generation and ground-truth benchmarking • Python/MATLAB for classification and carbon flux estimation	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-05

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc	X	
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-06

Student Proposed?	**Y/N**	N/A
ID	**RAV2026-06**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**A/Prof D Pillay (UCT Biological Sciences), G Maniel**	
Title	**RGB-D microtopography of sandprawn bioturbation**	
Description		

Background
Burrowing sandprawns are important ecosystem engineers on South African sandy shores (**Venter** et al., 2020). Species such as *Kraussillichirus kraussi* (Stebbing, 1900) constantly rework the sediment as they burrow and feed. They move sediment to the surface, form mounds and burrow openings, and change the physical structure and chemical properties of the sandflat (**Kristensen** et al., 2012). These changes affect the substratum itself and the communities that live in it (**Pillay** and Branch, 2011; **Thomas** et al., 2023). Sediment surface microtopography, like small mounds, burrow openings and roughness features, can be used as a proxy for macro-invertebrate ecological activity (**Schenone** et al., 2024, 2025). Measuring this surface structure therefore provides a practical way to quantify superficial bioturbation intensity. The difficulty is that these surfaces are fragile, and contact-based measurements can easily disturb the features being measured.

Low-cost RGB-D cameras, which record both colour and depth, offer a non-contact way to measure soft-sediment microtopography at the millimetre scale (**Azhar** et al., 2022; **Kamal** et al., 2014). **Azhar** et al. (2022) demonstrated a practical approach in which the sensor is shielded from sunlight, calibrated, and used to produce depth images of the sediment surface. A fitted polynomial surface is then removed from the depth image to isolate the small-scale topography, from which roughness metrics such as arithmetic mean roughness and rugosity can be calculated. A similar problem has been addressed at UCT for sea ice, where LiDAR and camera data were used to reconstruct and quantify the roughness of pancake ice (**Spirakis**, 2021; **Spirakis** et al., 2023). The same ideas of detrending, surface reconstruction and roughness extraction carry over well to soft-sediment microtopography.

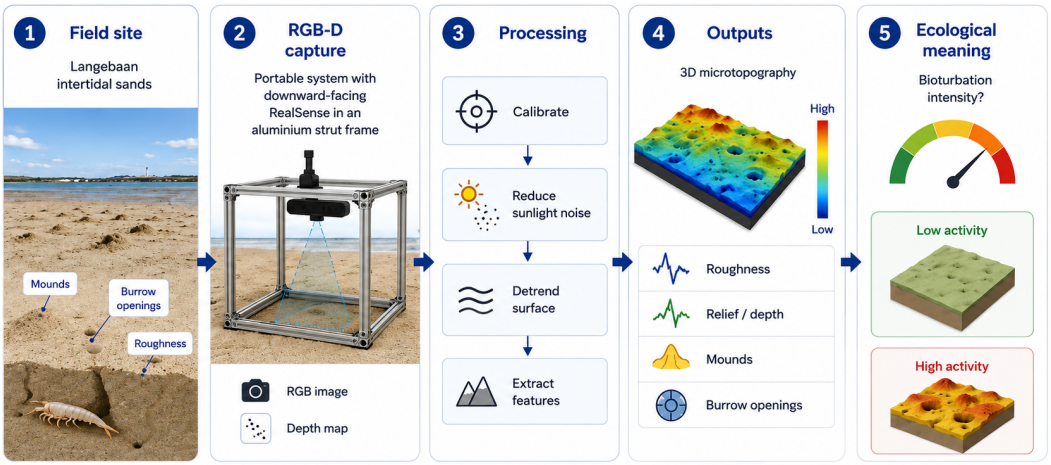


Fig. 6. Proposed RGB-D workflow for measuring sandprawn bioturbation. A downward-facing RealSense camera captures colour and depth images of the sediment surface, which are processed to recover microtopography, roughness, mounds and burrow openings. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation
This project develops an RGB-D microtopography pipeline to quantify sandprawn bioturbation, in collaboration with the Department of Biological Sciences at UCT. The work is based on the framework of **Azhar** et al. (2022), but adapted to the Intel RealSense camera. The main tasks are to calibrate the sensor, characterise its depth noise, and develop a processing pipeline that extracts the detrended sediment microtopography and the roughness metrics that describe surface disturbance. The system will be validated against controlled targets of known relief, so that the depth accuracy and roughness measurements can be quantified. As an extension, the project will also detect and segment the specific surface features produced by sandprawns, including mounds and burrow openings, using classical morphological image-processing methods. This allows the project to move beyond general roughness metrics and measure the visible bioturbation features directly.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of RGB-D microtopography measurement and sandprawn bioturbation, and critically evaluate the techniques that apply to this project. (c) To adapt and calibrate the Intel RealSense for microtopography capture, and characterise its depth noise under controlled laboratory conditions. (d) To develop the detrending and roughness-metric pipeline that extracts the microtopography from the depth images. (e) To detect and segment sandprawn features, the surface mounds and burrow openings, from the detrended depth. (f) To validate the pipeline in the laboratory against controlled targets of known relief, and characterise the geometric accuracy and sensitivity of the extracted microtopography metrics. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) An RGB-D microtopography pipeline on the Intel RealSense that extracts roughness metrics and detects sandprawn features, validated in the laboratory to quantify bioturbation intensity. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Image processing, 3D reconstruction, Interest in marine ecology
Extra Information	**References** M. **Azhar**, J. R. Hillman, T. Gee, S. Schenone, W. van der Mark, S. F. Thrush, and P. Delmas. An RGB-D framework for capturing soft-sediment microtopography. *Methods in Ecology and Evolution*, 13(8):1730–1745, 2022. **doi:** 10.1111/2041-210X.13899 S. **Kamal**, S. Y. Lee, and J. Warnken. Investigating three-dimensional mesoscale habitat complexity and its ecological implications using low-cost RGB-D sensor technology. *Methods in Ecology and Evolution*, 5(9):845–853, 2014. **doi:** 10.1111/2041-210X.12210 E. **Kristensen**, G. Penha-Lopes, M. Delefosse, T. Valdemarsen, C. Quintana, and G. Banta. What is bioturbation? the need for a precise definition for fauna in aquatic sciences. *Marine Ecology Progress Series*, 446:285–302, 2012. **doi:** 10.3354/meps09506 D. **Pillay** and G. M. Branch. Bioengineering effects of burrowing thalassinidean shrimps on marine soft-bottom ecosystems. *Oceanography and Marine Biology: An Annual Review*, 49:137–192, 2011. **doi:** 10.1201/b11009-5 S. **Schenone**, I. Bartl, and S. Thrush. Sediment microtopography predicts localised benthic ecosystem functioning and the effect of species interactions. *Marine Ecology Progress Series*, 749:47–55, 2024. **doi:** 10.3354/meps14735 S. **Schenone**, J. E. Hewitt, J. Hillman, R. Gladstone-Gallagher, J. Gammal, C. Pilditch, A. M. Lohrer, E. Ferretti, M. Azhar, P. Delmas, and S. F. Thrush. Seafloor sediment microtopography as a surrogate for biodiversity and ecosystem functioning. *Ecological Applications*, 35:e3069, 2025. **doi:** 10.1002/eap.3069 A. **Spirakis**. Investigation into surface roughness characterisation using LiDAR and cameras. BSc(Eng) final year project, Department of Electrical Engineering), University of Cape Town, 2021. **url:** https://tinyurl.com/spirakis2021 A. **Spirakis**, J. Hepworth, and R. Verrinder. 3D reconstruction of pancake sea ice using LiDAR and cameras. In *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pages 60–63. IEEE, 2023. **doi:** 10.1109/igarss52108.2023.10282250 C. M. **Thomas**, C. de Cerff, G. A. V. Maniel, A. E. Oyatoye, E. Rocke, H. G. Marco, and D. Pillay. Water filtration by endobenthic sandprawns enhances resilience against eutrophication under experimental global change conditions. *Scientific Reports*, 13(1):19067, 2023. **doi:** 10.1038/s41598-023-46168-y O. **Venter**, D. Pillay, and K. Prayag. Water filtration by burrowing sandprawns provides novel insights on endobenthic engineering and solutions for eutrophication. *Scientific Reports*, 10(1):1913, 2020. **doi:** 10.1038/s41598-020-58677-1
Broad Research Area	RGB-D imaging, 3D reconstruction, Microtopography, Marine ecology
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-06 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-06**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**A/Prof D Pillay (UCT Biological Sciences), G Maniel**	
Title	**RGB-D microtopography of sandprawn bioturbation**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in capturing and quantifying fine-scale sediment microtopography with a low-cost RGB-D device at the millimetre scale. It involves ill-posed and under-specified problems, such as the: • adaptation of the Intel RealSense to resolve millimetre-scale relief reliably on low-texture sediment, • separation of the sandprawn microtopography from the underlying sediment trend and sensor noise. It also presents high-level problems, such as: • Calibrating the device and characterising its depth noise under controlled laboratory conditions. • Detecting and segmenting sandprawn mounds and burrow openings from the detrended depth. • Relating the extracted metrics to sandprawn bioturbation intensity. The project also necessitates handling complex and incomplete information by: • Synthesising information from RGB-D imaging, surface metrology, and sandprawn ecology literature. • Balancing technical challenges, engineering design, and the interests of the Biological Sciences collaboration.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative implementation of solutions to: • Adapt, calibrate, and characterise the Intel RealSense for microtopography capture. • Develop the detrending and roughness-metric pipeline. • Detect and segment sandprawn features from the detrended depth. • Validate the pipeline in the laboratory against controlled targets of known relief. • Apply statistical analysis to evaluate the metrics as a bioturbation proxy.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be computational and experimental in nature and will require competencies in sensor calibration, depth-image processing, 3D reconstruction, and data analysis. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate image processing and data analysis tools to achieve project objectives. **Software required may include:** • Python/C++ with the Intel RealSense SDK for device interfacing and calibration • Python/MATLAB with OpenCV for depth-image processing and detrending • Python/MATLAB for feature detection, roughness metrics, and analysis	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-06

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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EEE4022S RAV2026-07

Student Proposed?	Y/N	N/A
ID	**RAV2026-07**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr MK Mason (MMRU, IDM)**	
Title	**Compact sealed stage-top incubator for long-duration live bacterial imaging**	

Description

Background

Mycobacteria are not uniform populations of simple rod-shaped cells. Even within a genetically identical culture, individual cells differ in growth rate, division timing, morphology, stress response and antibiotic susceptibility (**Aldridge** et al., 2012). These differences matter because mycobacterial behaviour is often expressed at the single-cell and microcolony scale: cells elongate, divide asymmetrically, form cords and organised aggregates, and interact with surfaces in ways that are lost in bulk culture measurements. Understanding these processes requires live imaging over time, at sufficient resolution to follow individual cells and the structures they form.

This need sits within a broader UCT mycobacterial imaging programme, where microscopy is used to study the physical and biological behaviour of mycobacteria, including cording, surface-associated growth and host-relevant infection models. In the longer term, this work links to contained imaging approaches for pathogenic mycobacteria and lung-relevant systems, including cell monolayers, Matrigel-supported interfaces and precision-cut lung slices. However, before such workflows can be translated into more specialised containment settings, the underlying imaging hardware must be robust, well characterised and safe to develop. A practical engineering step is therefore to build and validate the environmental control platform using a non-pathogenic model organism.

Long-duration time-lapse microscopy places strict demands on the sample environment. Temperature affects growth rate and division timing, small agarose-pad or chamber volumes are vulnerable to evaporation, and changes in humidity, gas composition or thermal gradients can alter the biology being measured (**Frigault** et al., 2009). The sample must remain on the microscope for hours to days, which means that incubation has to happen at the microscope stage itself. The system must maintain a stable environment while preserving optical access, avoiding condensation, and limiting focal-plane drift. If these conditions are not controlled, the resulting images may reflect stress caused by the imaging setup rather than the biological process under investigation.

Commercial stage-top incubators can provide temperature, humidity and gas control, but they are expensive, closed to modification, and usually designed for general mammalian cell culture rather than adaptable bacterial live-imaging workflows (**O'Carroll** et al., 2023; **Yan** et al., 2024). For this collaboration, the useful system is a compact, modifiable and well-characterised platform that can be built, repaired and adapted locally. The engineering challenge is therefore to design more than a heated box: the system must regulate temperature, humidity and carbon dioxide in a small sealed chamber, remain compatible with an inverted microscope, and provide stable imaging conditions for biologically meaningful bacterial time-lapse experiments.

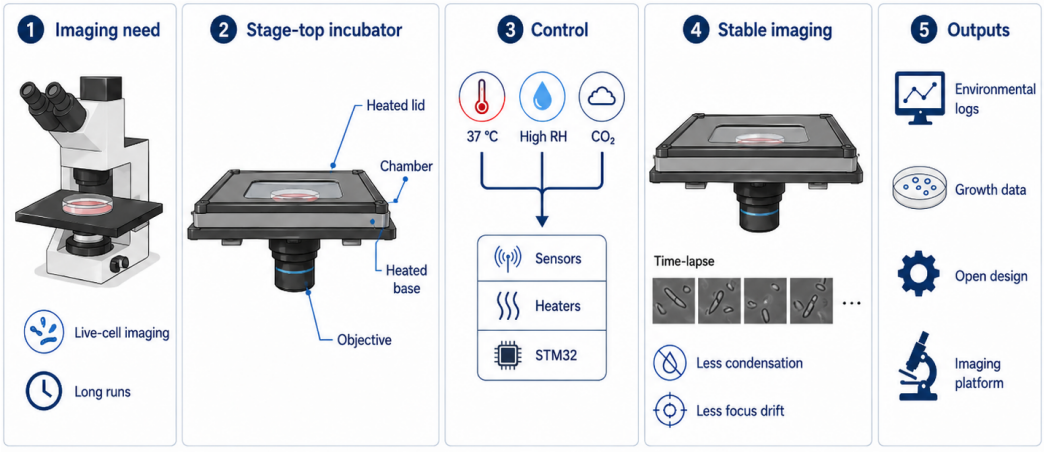


Fig. 7. Proposed stage-top incubator for long-duration bacterial imaging. A compact heated chamber maintains temperature, humidity and CO_2 around the sample while allowing time-lapse imaging on an inverted microscope. Schematic generated with assistance from ChatGPT (OpenAI, 2026) and edited by the author.

Subject of investigation

This project designs, builds and evaluates a compact sealed stage-top incubator for long-duration bacterial time-lapse imaging on an inverted microscope. The project will be developed as a safe engineering prototype using *Mycobacterium smegmatis*, a fast-growing, non-pathogenic mycobacterium that provides a useful model for testing live-imaging conditions, cell growth, morphology and microcolony behaviour without the containment requirements of pathogenic *M. tuberculosis* (**Sparks** et al., 2023; **Joyce** et al., 2011; **Aldridge** et al., 2012).

The incubator must maintain a stable internal environment near 37°C, with high humidity to reduce evaporation from small imaging volumes and controlled carbon dioxide where required. It must be compact enough to mount on a standard microscope stage, preserve optical access for time-lapse imaging, and address practical failure modes including condensation, uneven heating, thermal settling and focal-plane drift.

The internal conditions will be measured using digital temperature, humidity and carbon dioxide sensors, with an STM32-based controller used to drive the heaters and gas/humidity actuators, implement closed-loop control, and log the environmental data. The candidate will characterise the incubator in terms of setpoint tracking, stability, uniformity, repeatability, long-duration drift, condensation behaviour, image quality and focal stability.

The final system will be demonstrated on the microscope using live *M. smegmatis*, in consultation with the co-supervisor. The aim is to produce a working, characterised and modifiable stage-top incubator that supports current bacterial imaging while creating an engineering foundation for future mycobacterial imaging work, including cording, surface-associated growth and lung-relevant infection models, while keeping the student project within a safe development and validation scope.

Deliverables	**The key objectives of this project are:** (a) To understand the full scope and requirements of the project. (b) To conduct a literature review of previously published research in this field and critically evaluate current technologies and research that applies to this project. (c) To specify the operating point (a temperature near 37°C, a controlled near-saturated humidity and a defined carbon dioxide atmosphere) and the optical and mechanical constraints for mounting on a standard inverted microscope, and to select appropriate sensors and actuators. (d) To design and build the sealed chamber with heated optical windows, humidified gas path, and CO_2 mixing, within the footprint of a microscope stage. (e) To develop and implement closed-loop control and data-logging algorithms for the coupled multivariable system on an embedded platform. (f) To test and evaluate the system performance, characterising temperature stability and focus drift, condensation, carbon dioxide and humidity tracking, and uniformity across the imaging field, and to demonstrate imaging of a live *M. smegmatis* sample on the microscope. (g) To discuss the performance of the system, draw conclusions, and make recommendations for future improvements. **The key deliverables of this project are:** (a) A literature review report and detailed project statement and plan (Hand-in 2 weeks from the initiation of the project). (b) Weekly reporting of project progress and plans to the supervisor(s) through an agreed-upon communication platform, which could include online meetings. (c) A working prototype stage-top incubator that mounts on a standard inverted microscope and holds near 37°C, a controlled humidity and a defined carbon dioxide atmosphere under closed-loop control, with a host-side logging interface. (d) A research project report, following the Department of Electrical Engineering Final Year Project report guidelines. (e) Satisfactory completion of all Engineering Council of South Africa's Graduate Attributes (ECSA GAs). (f) A poster summarising the research project report. (g) A seminar to the Department summarising the research project report. (h) Attendance and presentation at the annual Department Open Day.
Skills/Requirements	Control systems, Embedded programming, Interest in instrumentation, optics and microbiology
Extra Information	**References** B. B. **Aldridge**, M. Fernandez-Suarez, D. Heller, V. Ambravaneswaran, D. Irimia, M. Toner, and S. M. Fortune. Asymmetry and aging of mycobacterial cells lead to variable growth and antibiotic susceptibility. *Science*, 335(6064): 100–104, 2012. **doi:** 10.1126/science.1216166 M. M. **Frigault**, J. Lacoste, J. L. Swift, and C. M. Brown. Live-cell microscopy — tips and tools. *Journal of Cell Science*, 122(6):753–767, 2009. **doi:** 10.1242/jcs.033837 G. **Joyce**, B. D. Robertson, and K. J. Williams. A modified agar pad method for mycobacterial live-cell imaging. *BMC Research Notes*, 4:73, 2011. **doi:** 10.1186/1756-0500-4-73 M. K. **Mason** and R. A. Verrinder. Biocontainment for live imaging of mycobacterial droplet evaporation: Uct Block Grant feasibility work. Molecular Mycobacteriology Research Unit, Institute of Infectious Disease and Molecular Medicine, UCT, 2024. Collaborative feasibility study R. **O'Carroll**, J. P. Reynolds, M. Al-Roqi, E. D. Aiyegbusi, and D. Dooley. ThermoCyte: an inexpensive open-source temperature control system for *in vitro* live-cell imaging. *Royal Society Open Science*, 10(11):231037, 2023. **doi:** 10.1098/rsos.231037 I. L. **Sparks**, K. M. Derbyshire, W. R. Jacobs, and Y. S. Morita. *Mycobacterium smegmatis*: the vanguard of mycobacterial research. *Journal of Bacteriology*, 205(1):e00337–22, 2023. **doi:** 10.1128/jb.00337-22 H. **Yan**, T. Wu, X. Li, Z. Feng, M. Ge, L. Zhang, and W.-F. Dong. Establishment of the microscope incubation system and its application in evaluating tumour treatment effects through real-time live cellular imaging. *Frontiers in Bioengineering and Biotechnology*, 12:1447265, 2024. **doi:** 10.3389/fbioe.2024.1447265 **Note:** the biological samples (*Mycobacterium smegmatis*, a non-pathogenic BSL-1 model organism) and the imaging protocol will be arranged with the co-supervisor, and any sterility or containment requirements will be handled through the co-supervisor's unit (MMRU, IDM).
Broad Research Area	Environmental control, Mechatronics, Embedded Systems, Biomedical instrumentation, Live-cell imaging
Project suitable for: ME/ECE/EE/ALL?	EE, EC, ME

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EEE4022S Topic RAV2026-07 GAs

Student Proposed?	Y/N	N/A
ID	**RAV2026-07**	
Supervisor	**RA Verrinder**	
Co-Supervisor(s)	**Dr MK Mason (MMRU, IDM)**	
Title	**Compact sealed stage-top incubator for long-duration live bacterial imaging**	
GA 1: Problem solving* Identify, formulate, analyse and solve complex engineering problems creatively and innovatively.	This project meets GA1 by addressing a complex engineering problem in the design and control of a sealed multivariable environment that must also function as part of an optical instrument. It involves ill-posed and under-specified problems, such as the: • specification of an achievable operating point and of acceptable stability, uniformity and focus-drift tolerances for imaging, • design of a controller for three coupled variables, temperature, humidity and CO_2, regulated by actuators with different time constants. It also presents high-level problems, such as: • Reconciling conflicting requirements, in particular a gas-tight seal to hold a defined carbon dioxide atmosphere against the objective access that imaging needs, and heated optical windows against the optical clarity and low condensation that imaging demands. • Achieving temperature stability tight enough to keep focus drift within micrometres over long imaging runs. • Managing the cross-coupling between the temperature, humidity and carbon dioxide loops, where actuating one variable disturbs the others. • Working within the footprint, mass and safety constraints of a microscope stage. The project also necessitates handling complex and incomplete information by: • Synthesising knowledge from thermal design, gas handling, optics, control theory and embedded systems, alongside the imaging requirements set by the co-supervisor, where no single standard solution exists. • Balancing technical performance, build cost and the needs of the researchers who will run imaging experiments in the chamber.	
GA 4: Investigations, experiments and analysis† Demonstrate competence to design and conduct investigations and experiments.	This project will require iterative investigation and experimentation to: • Identify the dynamic response of each subsystem, measuring step responses and the cross-coupling between the temperature, humidity and carbon dioxide loops. • Design and conduct experiments to quantify focus drift and condensation on the optics, and to tune and validate the closed-loop controllers against defined setpoints. • Map the uniformity of the internal environment across the imaging field. • Apply statistical analysis to quantify stability, repeatability and steady-state error, and to demonstrate a representative imaging run on the microscope.	
GA 5: Use of engineering tools‡ Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	This project will be both computational and practical in nature and will require competencies in embedded programming, control design, data handling and analysis, and mechanical, thermal and electronic design. As such, the candidate will demonstrate competency in the use of engineering tools and software, selecting appropriate tools to achieve the project objectives. **Software and tools required may include:** • C for embedded programming of the STM32 firmware, using the HAL • MATLAB/Python for control design, system identification and simulation • MATLAB/Python for data processing, analysis and validation, including image-based quantification of focus drift • CAD for the stage-top mechanics, optical mounts and the gas and moisture handling paths • Circuit design tools for the heater, valve and sensor interface electronics	

*Graduate Attribute 1: Problem solving

Complex engineering problems require in-depth fundamental and specialised engineering knowledge and have one or more of these characteristics:

- are ill-posed, under- or over-specified, or require identification and refinement;
- are high-level problems including component parts or sub-problems;
- are unfamiliar or involve infrequently encountered issues;

and their solutions have one or more of the characteristics:

- are not obvious, require originality or analysis based on fundamentals;
- are outside the scope of standards and codes;
- require information from a variety of sources that is complex, abstract or incomplete;
- involve wide-ranging or conflicting issues: technical, engineering and interested or affected parties.

†Graduate Attribute 4: Investigations, experiments and data analysis

The balance of **investigation and experiment** should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

An **investigation differs from a design** in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

‡Graduate Attribute 5: Engineering methods, skills and tools, including information technology

A range of methods, skills and tools appropriate to the disciplinary designation of the program including:

- discipline-specific tools, processes or procedures;
- computer packages for computation, modelling, simulation, and information handling;
- computers and networks and information infrastructures for accessing, processing, managing, and storing information to enhance personal productivity and teamwork.

Ethics clearance questionnaire: RAV2026-07

Question	Question Text	Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined **here**.		X
Answer is **No** here if your project does not utilize ML and AI			
Q5	Does this project involve external parties, funders, etc		X
Answer the following questions if you answer **Yes** to any of the above questions.			
If the answer is **Yes** to **Q1**, please answer the following questions:			
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X
If the answer is **Yes** to **Q2**, please answer the following questions:			
Q7	Are the third-party data used anonymous (data does not contain human or animal-related information?)		X
Q8	Are the third-party data used from an open source?		X
Q9	Are the third-party data used from a different research group?		X
Q10	If the answer to **Q9** is **Yes**, do you have the approval to use third-party data sets? Attach the proof to PSQ application.		X
If the answer is **Yes** to **Q5**, please answer the following questions:			
Q11	Have you signed an MOU between the parties [If Yes, attach the proof to PSQ application.]		X
Q12	Will there be a chance for any conflict of interest between the parties? [If **Yes**, provide details of the issue and your plan to solve it]		X

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